

LECTURE NO. 4 & 5

INTRODUCTION TO HAZARDS

Hazard Profiling, Vulnerability, Capacity, and Risk

Fall 2026



Learning Objective

After the completion of this lecture, students will be able to

- **UNDERSTAND** - the components of hazard and disaster risk management
- **LEARN** - Disaster Risk Management cycle -old and new concepts

We used many terms for disaster management, for example;

- Disaster
- Hazards
- Vulnerability
- Capacity
- Sensitivity
- Risk and Disaster Management

Now we know

- What is a Hazard?
- What is a Disaster?

Today we will discuss

- What is Hazard profiling?
- With example differentiate between vulnerability and capacity
- How disasters can be defined in terms of hazard, vulnerability, and capacity
- How can you relate poverty with vulnerability and capacity?
- Examples of Disasters and Disaster Management from ancient history?

"Hazards are always prevalent, but a hazard becomes a disaster only when the frequency or likelihood of a hazard and the vulnerability of the community increases the risk of being severely affected"

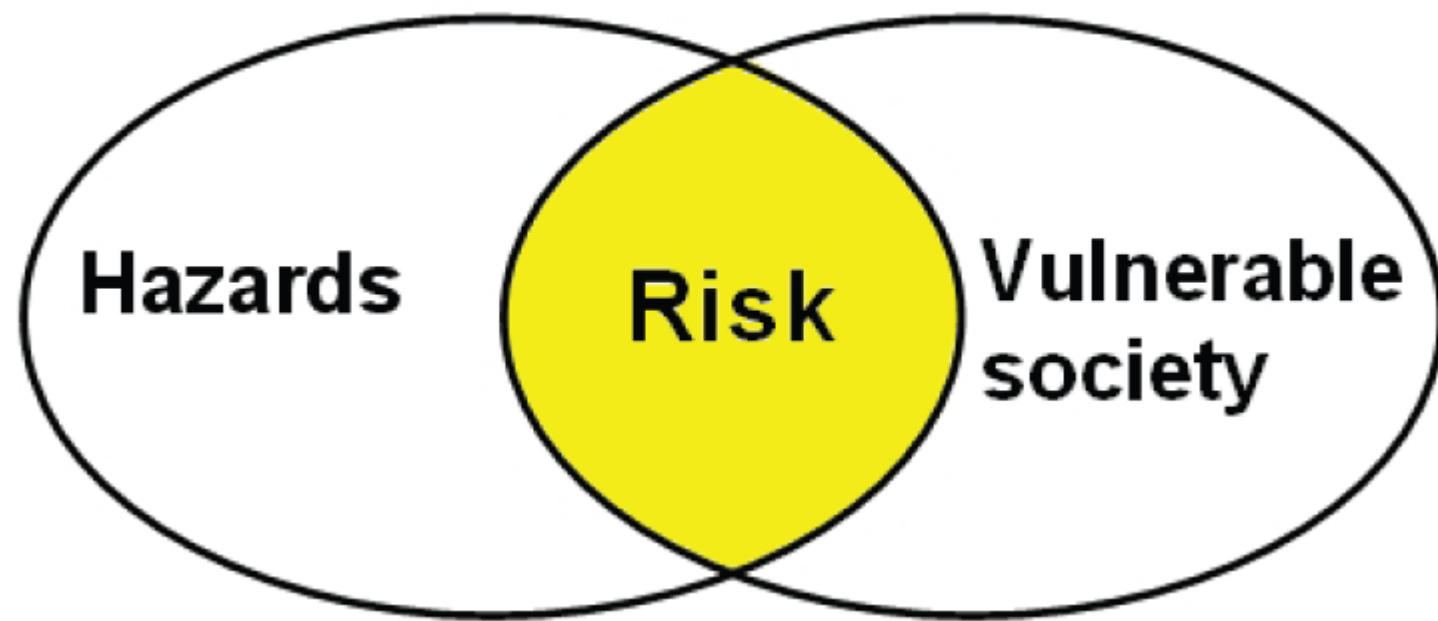
What is Risk?

Hazard and Risk

- Each Hazard carries an associated Risk.
- Risk depends on likelihood of a hazard leading to an actual disaster and the consequences of that disaster.
- When the response requirements for an emergency event exceed the capabilities of available emergency services the event is classified as a Disaster.
- Knowledge of Risk is essential for Preparedness, Planning and Mitigation activities.

Risk : Definition

- Measure of the expected losses due to a hazard event occurring in a given area over a specific time period.
 - Function of the probability of particular hazardous event and the losses each would cause.
- The potential disaster losses, in lives, health status, livelihoods, assets and services, which could occur to a particular community or a society over some specified future time period (UNISDR).

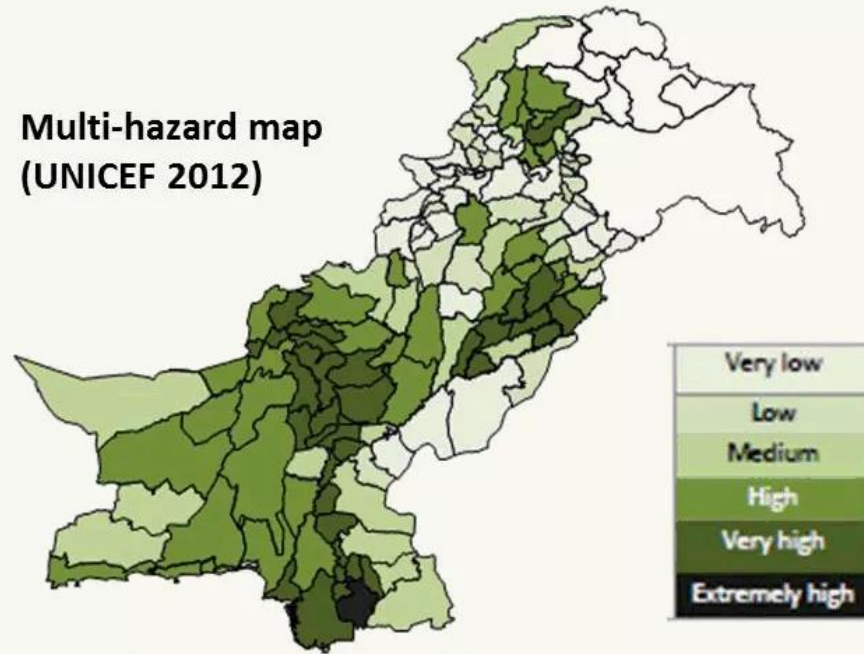


Risk depends on

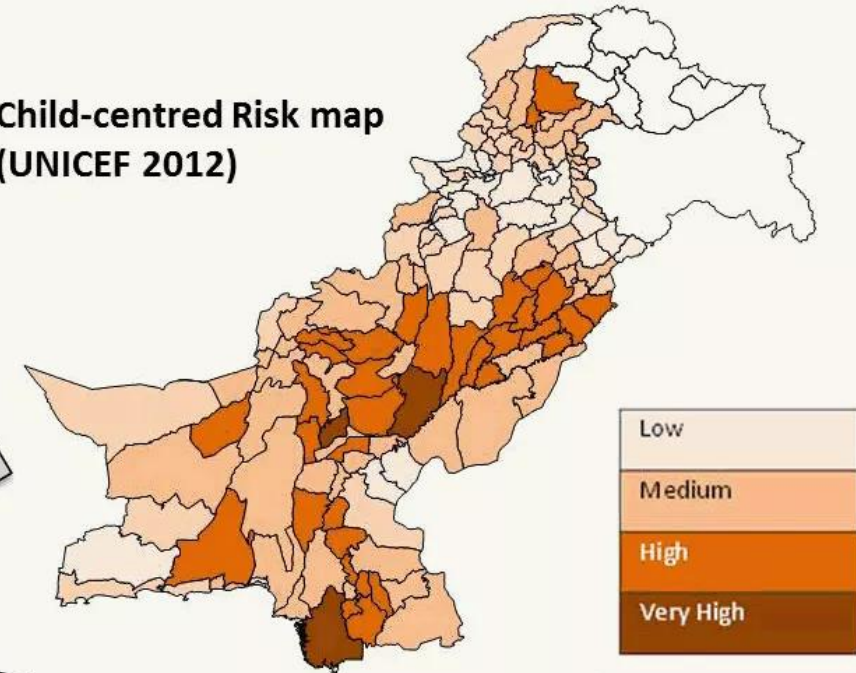
- Nature of the hazard
- Vulnerability of the elements which are affected
- Capacity to recover
- Economic value of those elements

Disaster risk management includes all measures which reduce disaster related losses of life, property, or assets by either reducing the hazard or vulnerability of the elements at risk.

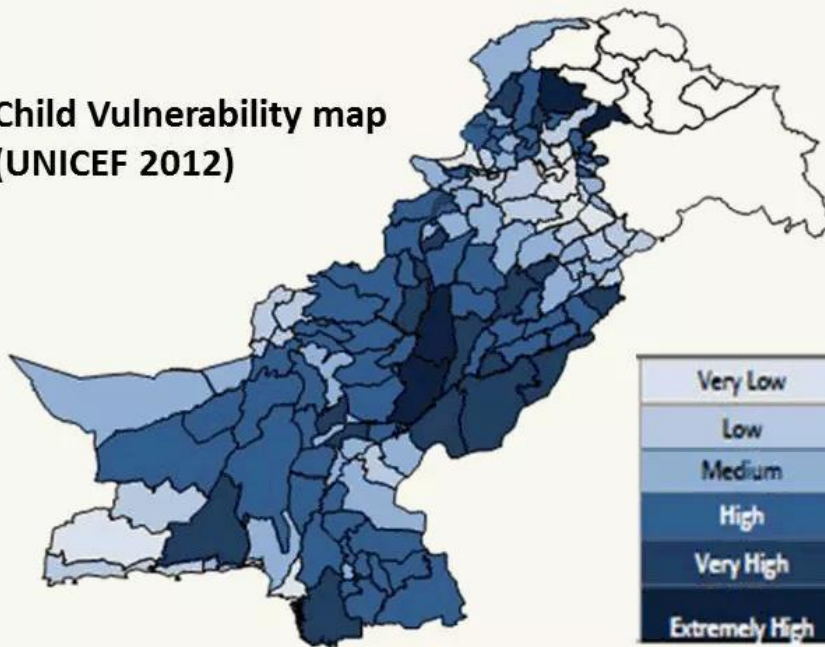
**Multi-hazard map
(UNICEF 2012)**



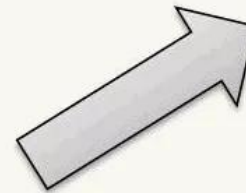
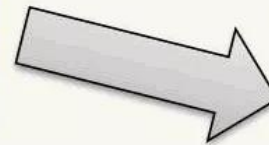
**Child-centred Risk map
(UNICEF 2012)**

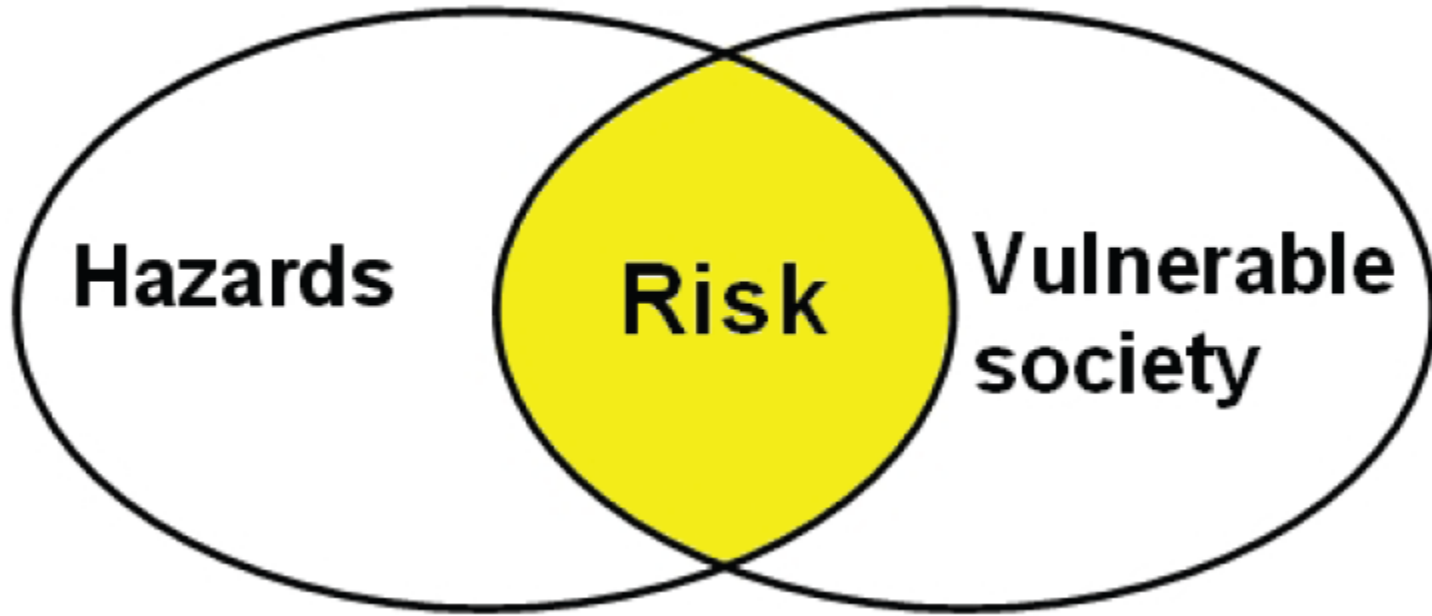


**Child Vulnerability map
(UNICEF 2012)**



**Exposure
(child population)**





When Hazard and risk materialize
then Disaster occurs



Vulnerability?

The extent to which a community, structure, services or geographic area is likely to be damaged or disrupted by the impact of a particular hazard, on account of their nature, construction, and proximity to hazardous terrains or a disaster prone area.

Vulnerability?

- Physical Vulnerability

- who and what may be damaged or destroyed by a natural hazard

- Socio Economic Vulnerability

- socio-economic condition of the people affected by a disaster

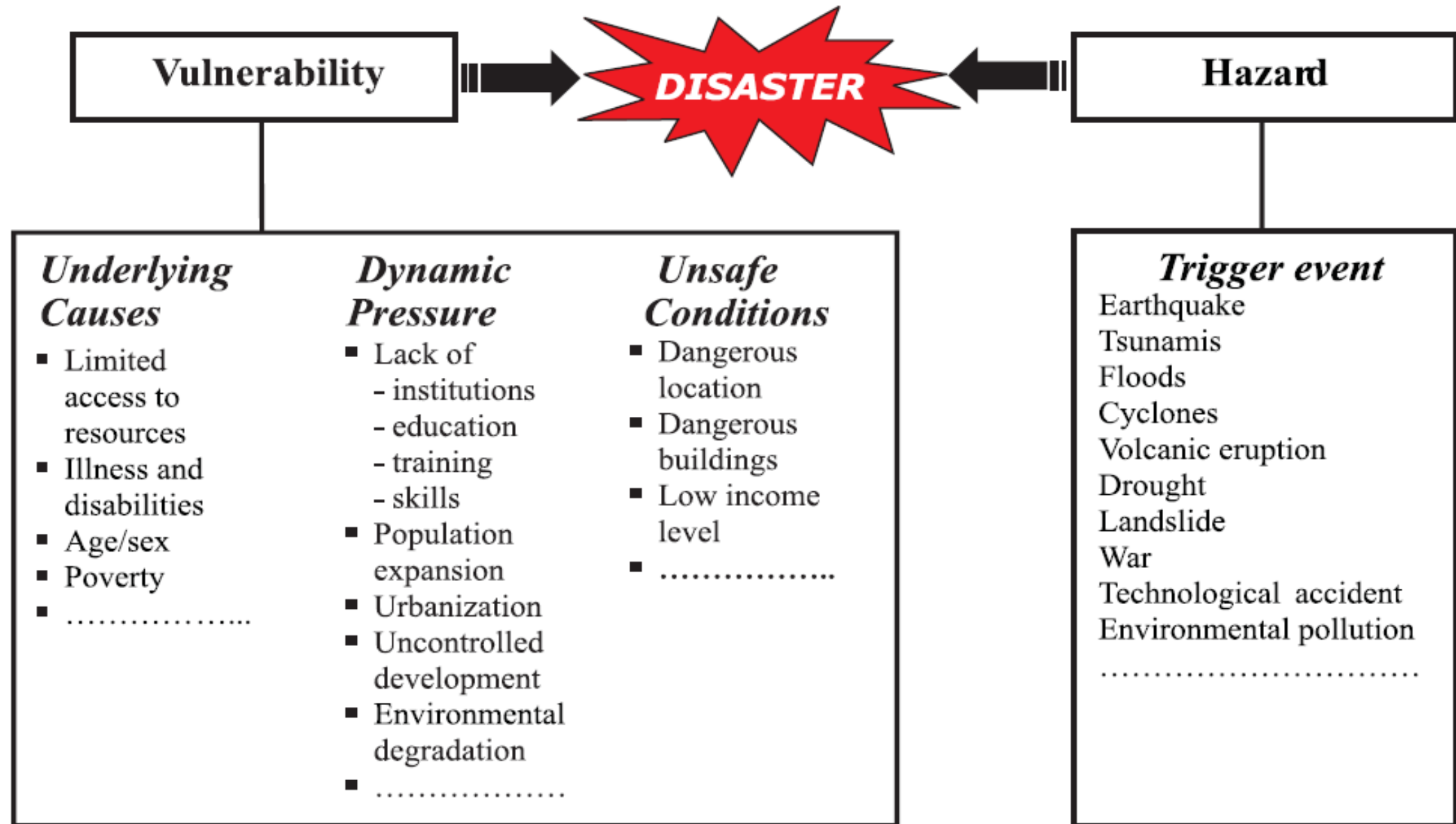


Poverty --
vulnerability

Vulnerability Examples?

- Share Examples?
 1. poor design and construction of buildings
 2. inadequate protection of assets
 3. lack of public awareness
 4. limited recognition of risks and preparedness measures and lack of proper management

Disaster occurs when hazard and vulnerability meet

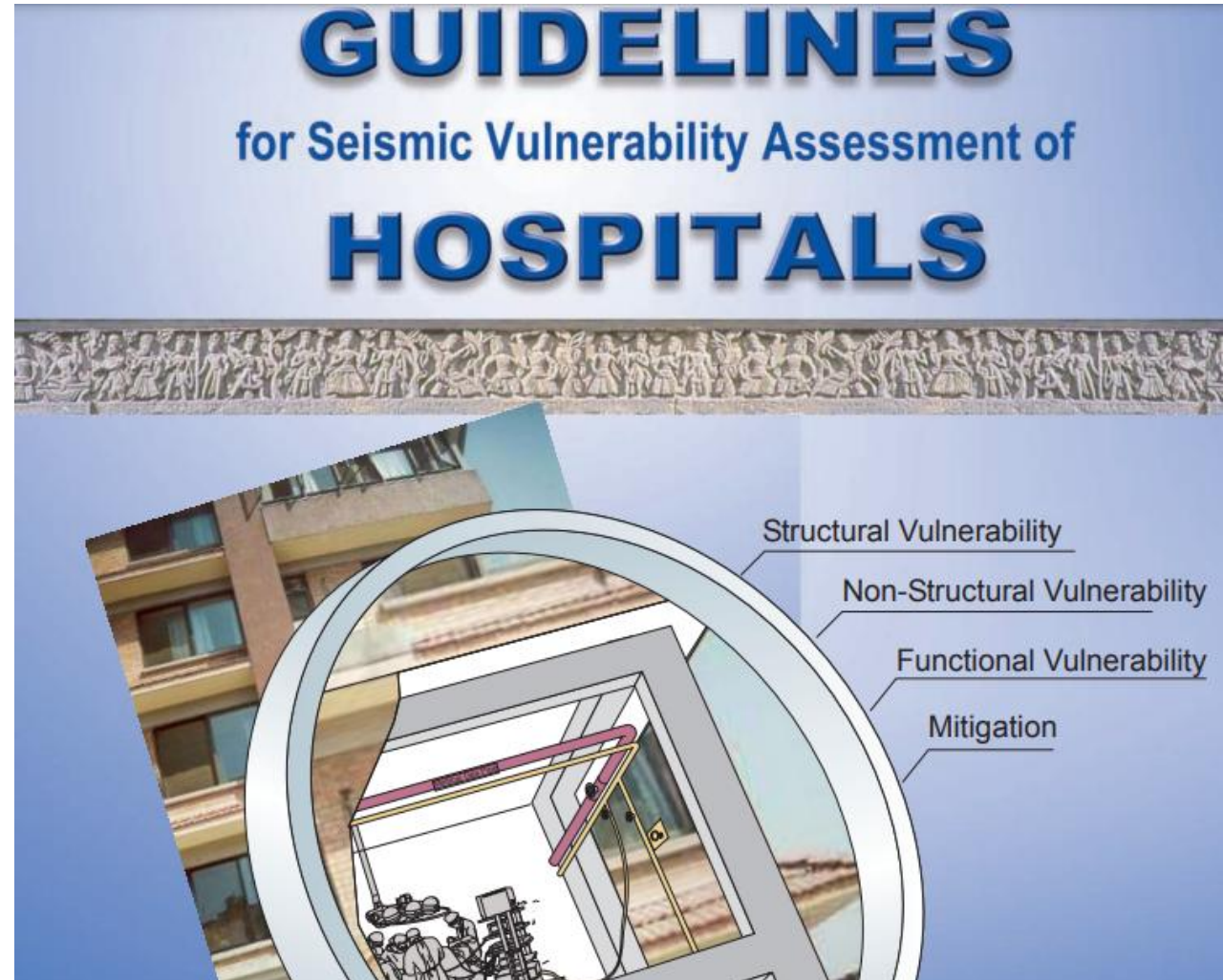


EFFECTS OF DISASTERS

	Human - social	Physical	Economic
Primary Effect	• Fatalities • Injuries • Loss of income or employment opportunities • Homelessness	• Structural damage or collapse to buildings and infrastructure • Non-structural damage and damage to contents • Ground deformation or loss of ground quality	• Interruption of business due to damage to buildings and infrastructure • Loss of productive workforce through fatalities, injuries and relief efforts • Capital costs of response and relief
Secondary Effect	• Disease • Permanent disability • Psychological impact • Loss of social cohesion due to disruption of community • Political unrest (government response is perceived as inadequate)	• Progressive deterioration of damaged buildings and infrastructure which are not repaired	• Loss of markets and trade opportunities through short-term business interruption • Capital costs of repair • Loss of confidence by investors, withdrawal of investment • Losses borne by the insurance industry weakening the insurance market and increasing premiums

Vulnerability assessment

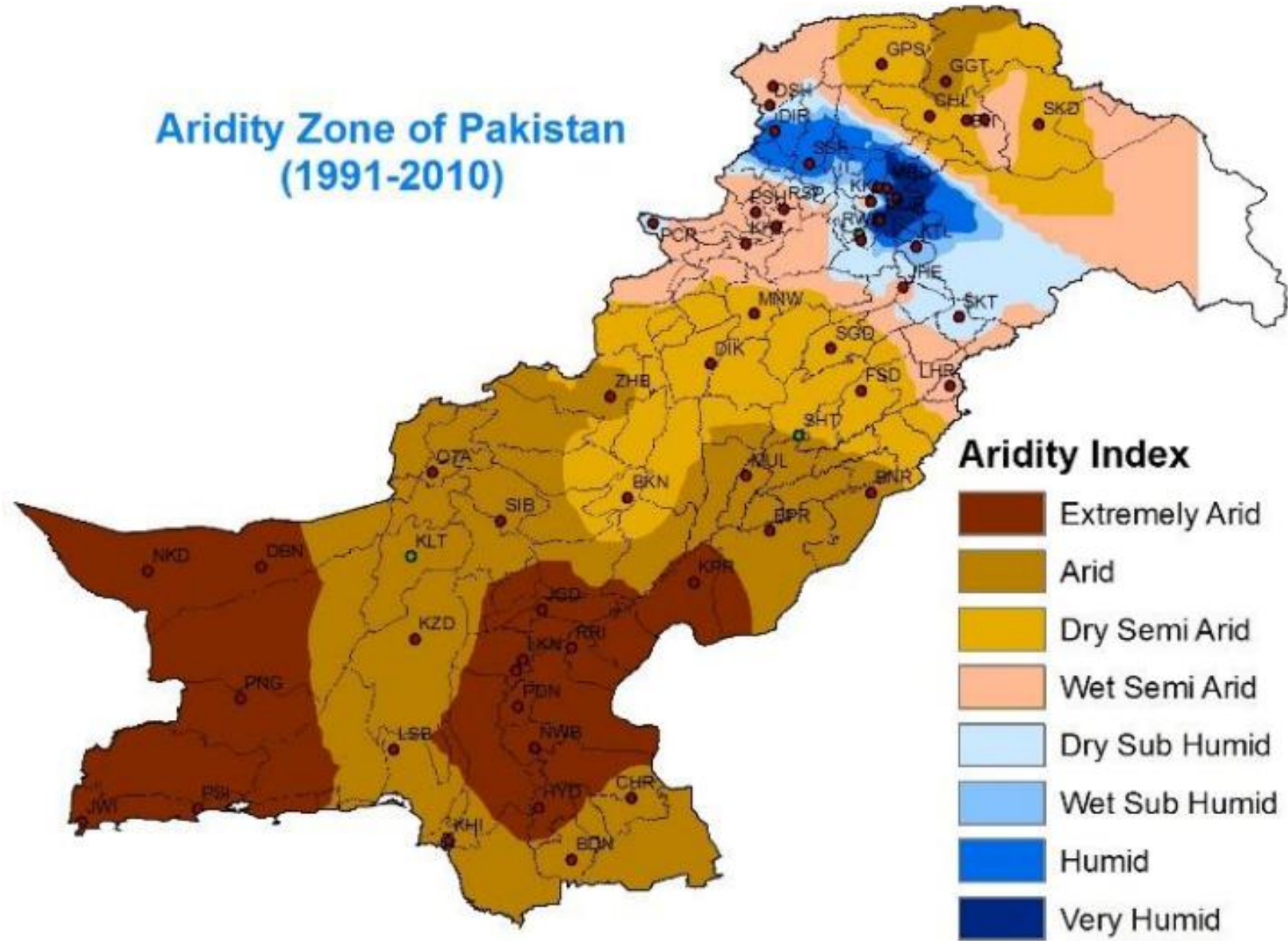
- Different for different
 - Hazards
 - Element at risk
 - Environment/conditions/location
 - Etc.



Vulnerability Maps

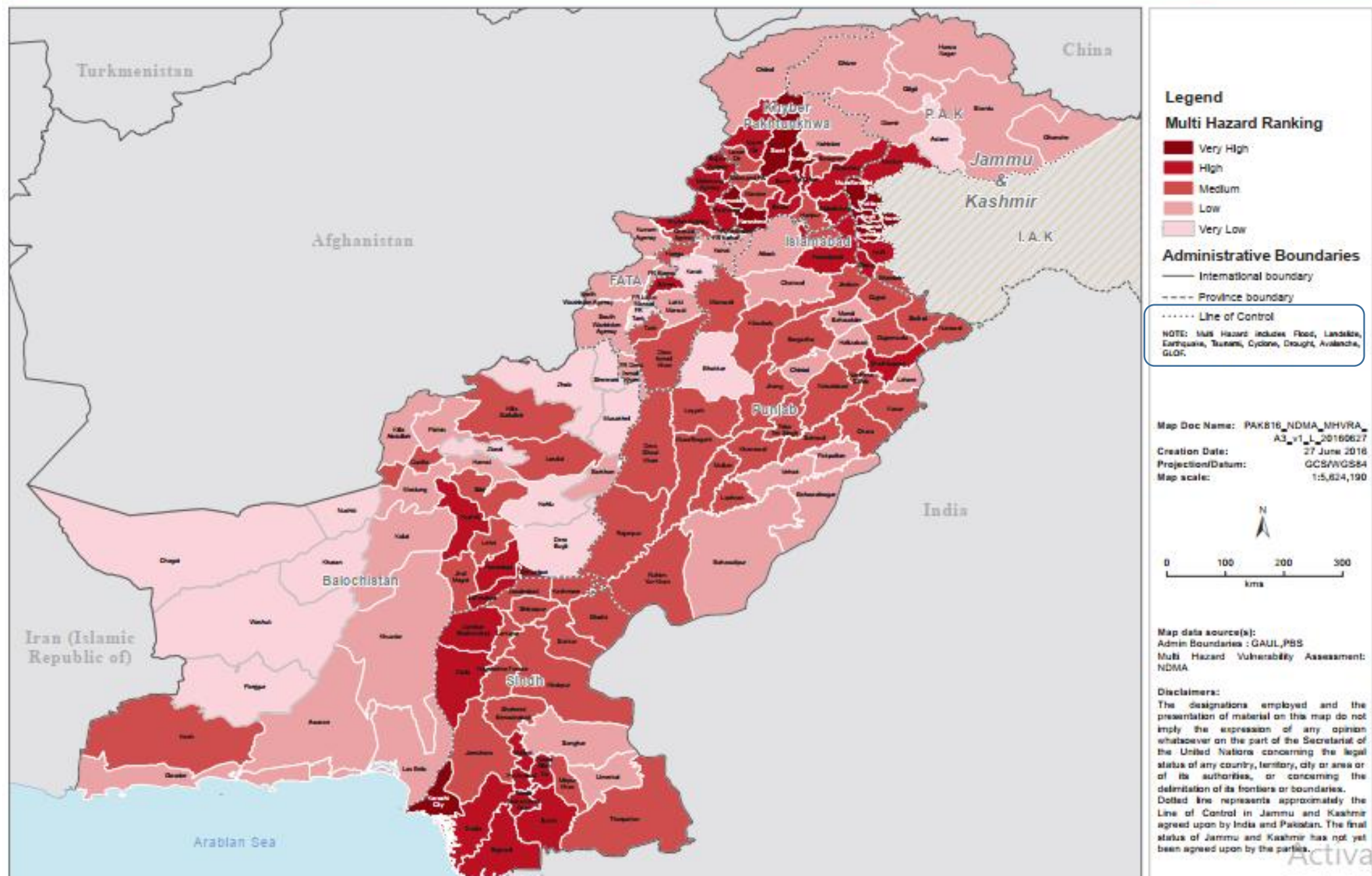
Examples

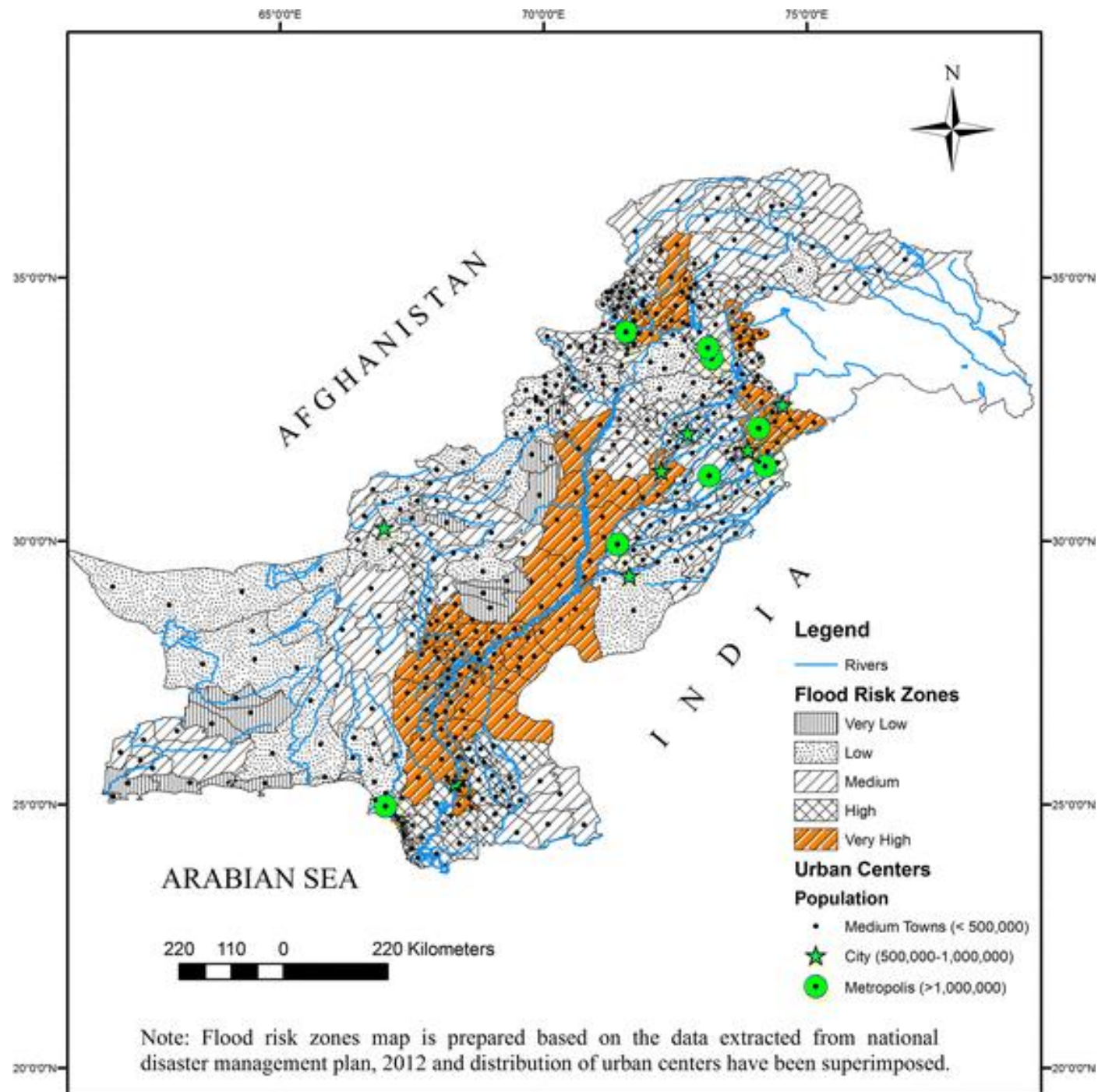
Aridity Zone of Pakistan (1991-2010)



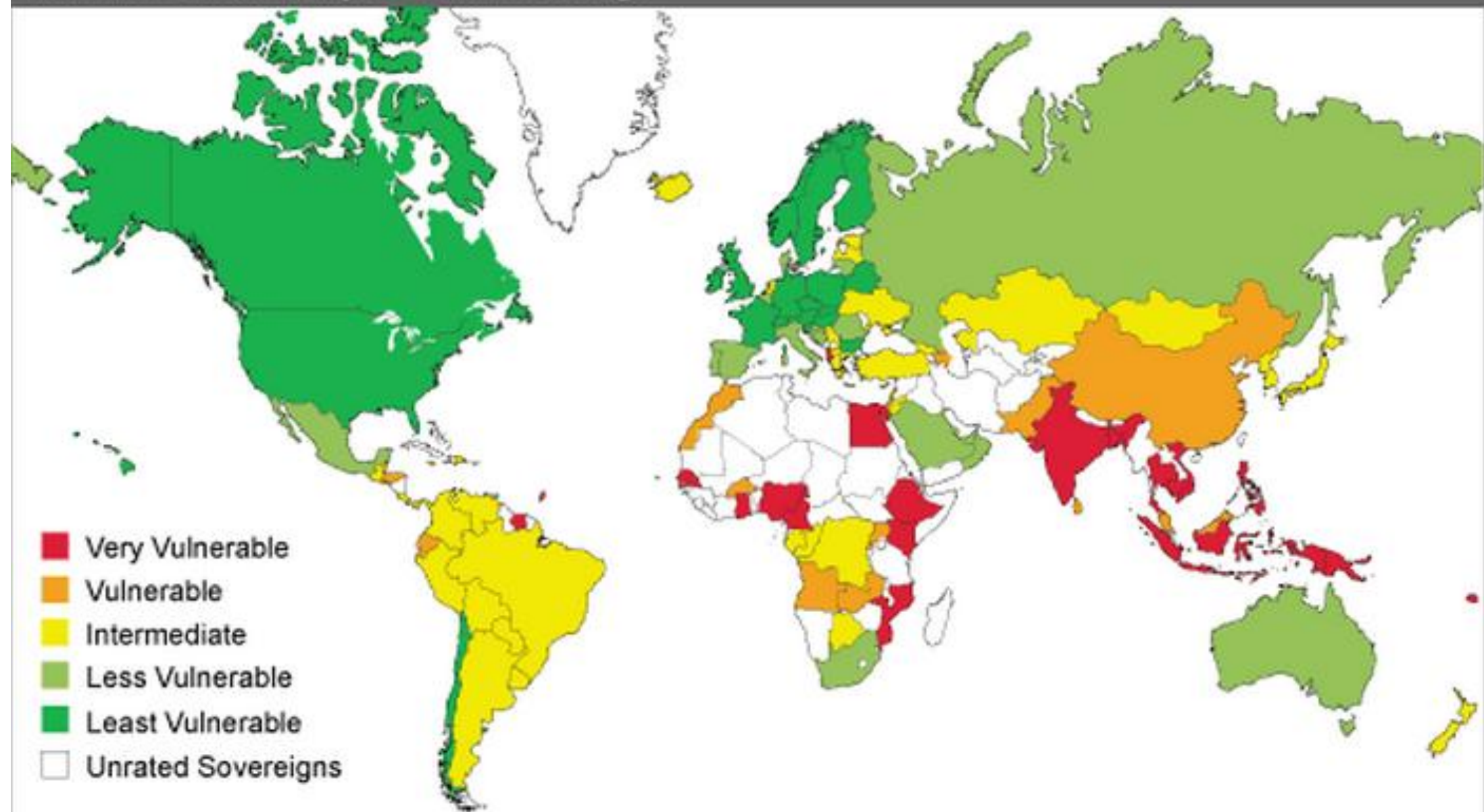
Source: Pakistan Meteorological Department (2000)

PAKISTAN: Multi- Hazard Vulnerability and Risk Assessment (MHVRA)





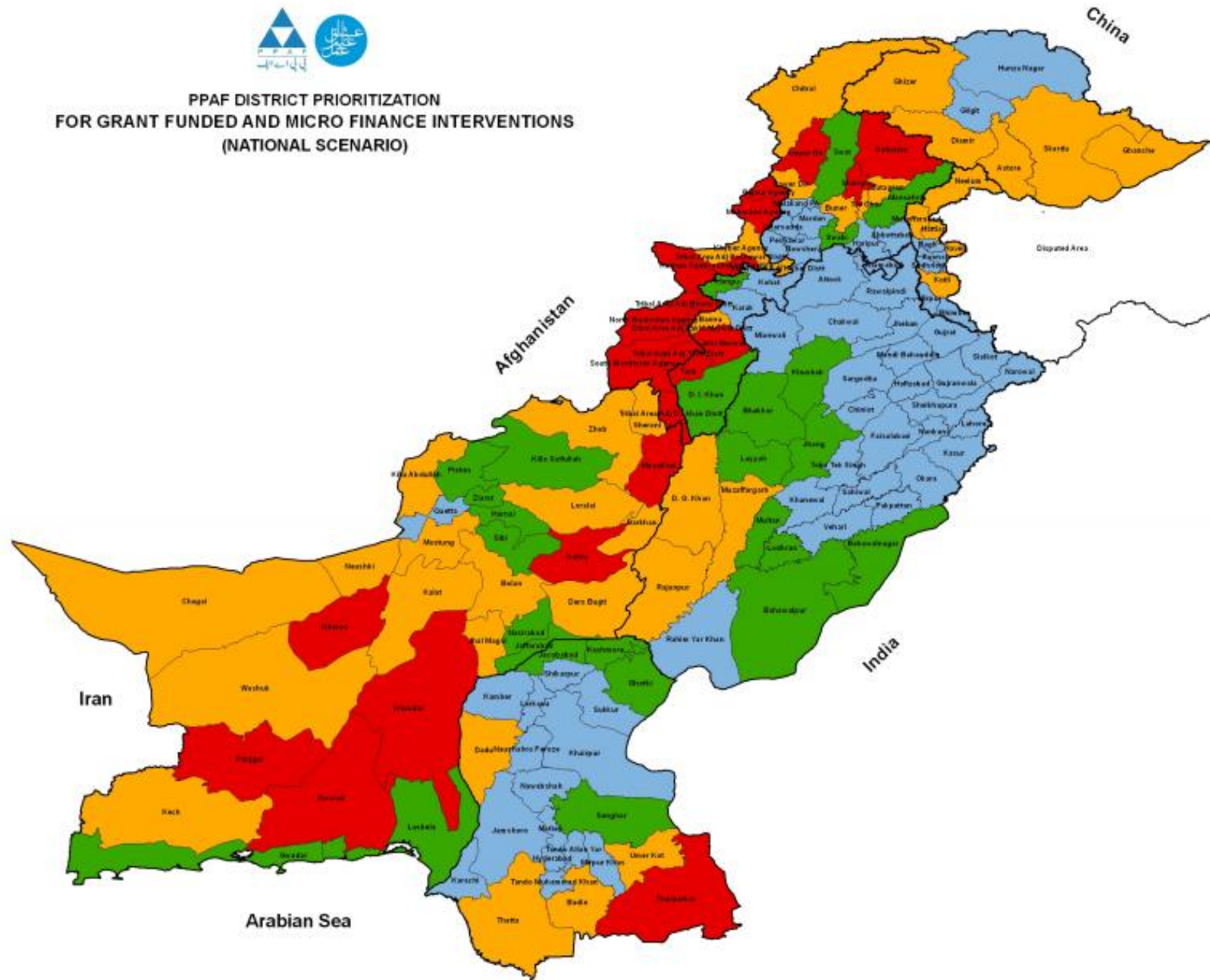
Potential Vulnerability To Climate Change



© Standard & Poor's 2014.



**PPAF DISTRICT PRIORITIZATION
FOR GRANT FUNDED AND MICRO FINANCE INTERVENTIONS
(NATIONAL SCENARIO)**



Note:

1. PPAF grant funding can be directed to all districts, other than those shown in blue color. Interventions in latter type of districts will be supported through Micro Finance.

**Priority One:
Lowest HDI Index and Food Security**

1. Awaran
2. Khuzdar
3. Kohlu
4. Musakhel
5. Panjgur
6. Kohistan
7. Lakki Marwat
8. Shangla
9. Tank
10. Upper Dir
11. Tharparkar
12. Kharan
13. North Waziristan Agency
14. South Waziristan Agency
15. Bajaur Agency
16. Kurum Agency
17. Mohmand Agency
18. Tribal Area Adj. Lakki Marwat Distt
19. Orakzai Agency
20. Tribal Area Adj. Bannu Distt
21. Tribal Area Adj. D.I. Khan Distt
22. Tribal Area Adj. Tank Distt

**Priority Two:
Low HDI Index and Food Security**

1. Bolan
2. Chagai
3. Khyber Agency
4. Kalat
5. Kech
6. Killa Abdullah
7. Kotli
8. Loralai
9. Mastung
10. Naushahi
11. Astore
12. Skardu
13. Diamir
14. Neelum
15. Ghazne
16. Ghizer
17. Thatta
18. Dera Bugti
19. Buner
20. Badin
21. Chitral
22. Tribal Area Adj. Peshawar Distt
23. Dadu
24. Umer Kot
25. Washuk
26. Zhob
27. Sherani
28. Haveli
29. Barikhan
30. Hattian
31. Muzaffargarh
32. Jhal Magsi
33. Bannu
34. Batagram
35. D. G. Khan
36. Muzaffargarh
37. Rajapur
38. Lower Dir
39. Tribal Area Adj. Kohat Distt
40. Tor Ghar

**Priority Three:
Moderate HDI Index and Food Security**

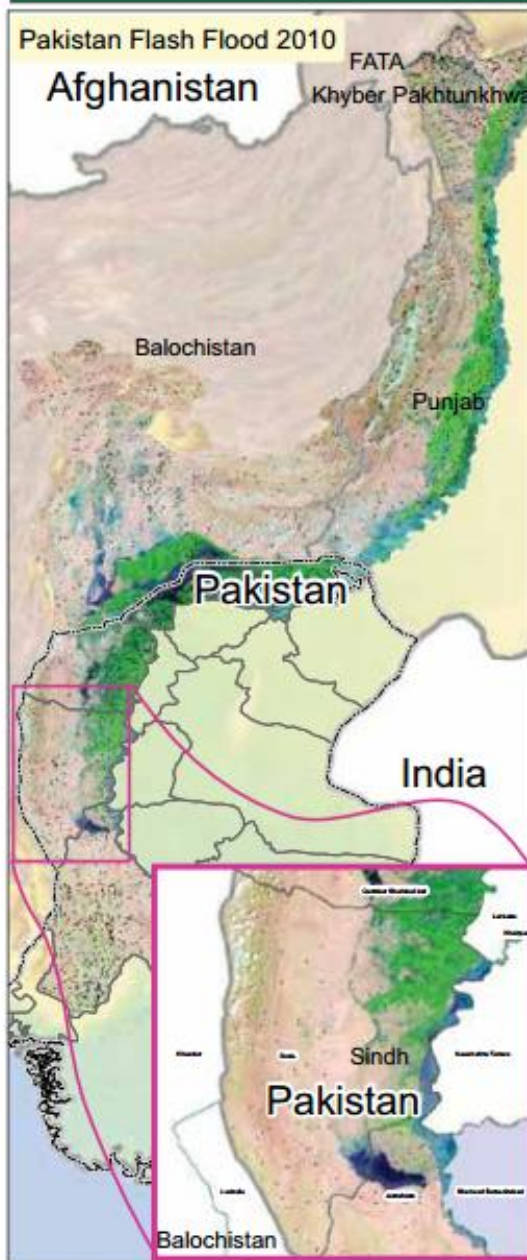
1. Gwadar
2. Larkana
3. Ziarat
4. Pishin
5. Jhang
6. Killa Saifullah
7. D. I. Khan
8. Hange
9. Ghotki
10. Swabi
11. Sui
12. Jacobabad
13. Kashmore
14. Sibi
15. Hamai
16. Jaffarabad
17. Nasirabad
18. Sanghar
19. Bahawalnagar
20. Bahawalpur
21. Bhakkar
22. Khushab
23. Layyah
24. Lodhran
25. Multan
26. Rahim Yar Khan
27. Manshera

**Priority Four:
Good HDI Index and Food Security**

1. Bagh
2. Bimber
3. Mirpur
4. Poonch
5. Sudhnoti
6. Quetta
7. Chiniot
8. Islamabad
9. Mianwali
10. Abbottabad
11. Charsadda
12. Haripur
13. Karak
14. Kohat
15. Malakand PA
16. Mardan
17. Nowshera
18. Peshawar
19. Attock
20. Chakwal
21. Faisalabad
22. Gujranwala
23. Gujrat
24. Hafizabad
25. Jhelum
26. Kasur
27. Khanewal
28. Lahore
29. Mandi Bahaudin
30. Nankana
31. Narowal
32. Okara
33. Pakpattan
34. Rawalpindi
35. Sahiwal
36. Sargodha
37. Sheikhupura
38. Sialkot
39. Toba Tek Singh
40. Vehari
41. Hyderabad
42. Jamshoro
43. Kamber
44. Karachi
45. Khairpur
46. Larkana
47. Mafiar

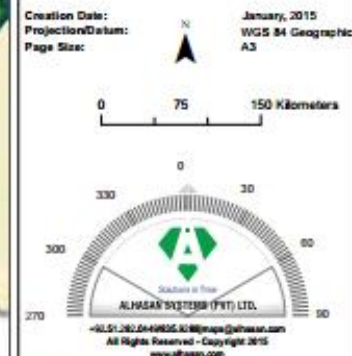
Sindh - Dadu District Flash Flood Vulnerability Map from Year-2010 to Year-2013

Date January 2015



Legend

- Settlements
- Flood Water
- Soil
- Vegetation Cover
- Clouds Cover
- District Boundary
- Provincial Boundary
- International Boundary



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Vulnerability

IPCC definition (Turner II et al., 2003)

- is a function of
 - Exposure
 - Sensitivity (of a system to changes in climate and the ability to adapt to system to changes in climate)
 - Adaptive capacity

Vulnerability: Several definitions and explanations

- Look at
- <https://www.weadapt.org/knowledge-base/vulnerability/vulnerability-definitions>

Box 1. Common definitions of vulnerability in the climate change field.

IPCC

The propensity or predisposition to be adversely affected. Vulnerability encompasses a variety of concepts and elements including sensitivity or susceptibility to harm and lack of capacity to cope and adapt. (IPCC Glossary, 2014, p. 128).

UNISDR

The conditions determined by physical, social, economic and environmental factors or processes which increase the susceptibility of an individual, a community, assets or systems to the impacts of hazards. (UNISDR, 2016, p. 24).

Vulnerability definitions

Submitted by Sukaina Bharwani | published 25th Mar 2011 | last updated 4th Nov 2020

The ordinary use of the word 'vulnerability' refers to the capacity to be wounded, i.e., the degree to which a system is likely to experience harm due to exposure to a hazard.

The scientific use of 'vulnerability' has its roots in geography and natural hazards research but this term is now a central concept in a variety of research contexts such as natural hazards and



Vulnerability Reduction

- Developing "disaster resistant" communities

Any Idea - How?

- How can we design and build our communities in the first place in these hazard-prone areas?

Capacity

"Resources, means, and strengths which exist in communities to enable them to cope with, withstand, prepare for, prevent, mitigate, or quickly recover from a disaster"

Capacity to Respond and Recover?

- Community to respond to and recover from a hazard in order to reduce its potential impact
- The positive managerial capabilities of individuals, households, and communities to confront the threat of disasters (e.g. through awareness raising, early warning, and preparedness planning)

IRMA

GO

AWAY!

9.9.2017

Capacity to Respond and Recover?

- Physical Capacity

- Capacity to salvage things after disasters or skills that may lead to new opportunities.

- Socio Economic Capacity

- Capacity to recover soon because of their good socio-economic conditions.

Capacity: Examples

- Please share few examples from your daily life

Terminology by UN International Strategy for Disaster Reduction (UNISDR)

Risk: The combination of the probability of an event and its negative consequences.

Acceptable risk: The level of potential losses that a society or community considers acceptable given existing social, economic, political, cultural, technical and environmental conditions.

Disaster risk: The potential disaster losses, in lives, health status, livelihoods, assets and services, which could occur to a particular community or a society over some specified future time period.

Capacity: The combination of all the strengths, attributes, and resources available within a community, society, or organization that can be used to achieve agreed goals.

- **Vulnerability:** The characteristics and circumstances of a community, system, or asset that make it susceptible to the damaging effects of a hazard.
- **Exposure:** People, property, systems, or other elements present in hazard zones that are subject to potential losses.
- **Forecast:** Definite statement or statistical estimate of the likely occurrence of a future event or conditions for a specific area.

Comment: In meteorology a forecast refers to a future condition, whereas a warning refers to a potentially dangerous future condition.

Prevention: The outright avoidance of adverse impacts of hazards and related disasters.

- **Mitigation:** The lessening or limitation of the adverse impacts of hazards and related disasters.
- **Preparedness:** The knowledge and capacities developed by governments, professional response and recovery organizations, communities and individuals to effectively anticipate, respond to, and recover from, the impacts of likely, imminent, or current hazard events or conditions.
- **Response:** The provision of emergency services and public assistance during or immediately after a disaster in order to save lives, reduce health impacts, ensure public safety, and meet the basic subsistence needs of the people affected.
- **Recovery:** The restoration, and improvement where appropriate, of facilities, livelihoods, and living conditions of disaster-affected communities, including efforts to reduce disaster risk factors.

- **Resilience:** The ability of a system, community or society exposed to hazards to resist, absorb, accommodate to and recover from the effects of a hazard in a timely and efficient manner, including through the preservation and restoration of its essential basic structures and functions.

Difference between a disaster
and
an Emergency?

Disaster Characteristics (Dr. Cees Van Westen)

An extreme phenomenon

- of large intensity (e.g. a measurable quantity such as earthquake intensity, water depth)
- of limited duration (which can vary from seconds to months, but should be defined in time);
- occurring at a certain location (this spatial component will be very important in this course);
- involving a complex interplay between physical and human systems;
- causing
 - loss of lives and threats to public health, as well as physical damage
 - disruption of livelihood systems and society;
- **exceeding local capacities and resources and requiring outside assistance to cope with**

Emergency?

- Disaster is an emergency in which local authorities can not cope with the impact or scale of the hazard, and therefore, the event is managed outside the affected communities.
 - Earthquakes, Hurricanes, Floods, etc.
- Non-disaster Emergencies
 - Local floods, local epidemics, transport or industrial accidents, environmental pollution etc.

Hazard, Emergency and Disaster

		Time/probability		
		Uncertain	Imminent	Occurred
Demand compared to community capacity	Less than	Hazard	Emergency	Emergency
	Greater than	Hazard	Emergency	Disaster/catastrophe

Source: FEMA

Define which of the following is a disaster?

1. When you become ill, and cannot work anymore?
2. When a famous football player is injured and misses the most important match, and his team loses the world championship?
3. When does a car accident become a disaster? The annual financial cost of car accidents in the US is estimated to be around 230 Billion dollars, with 2.9 million injuries and around 43,000 casualties.
4. The death of 2,974 people in the attack on the Twin Towers on 9/11/2001?
5. The big coronavirus (COVID19) that hit the world?

Homework-2

Answer the following:

- What are the legal definitions used by FEMA (USA) to declare a major disaster?
- How do insurance companies differentiate between standard accidents and catastrophic events?
- How does NDMA define disasters?

Hazard Profiling

Identify all hazards that have affected or could affect the area of focus through

- Historical records
- Brainstorming
- Scientific analysis
- Subject matter expertise

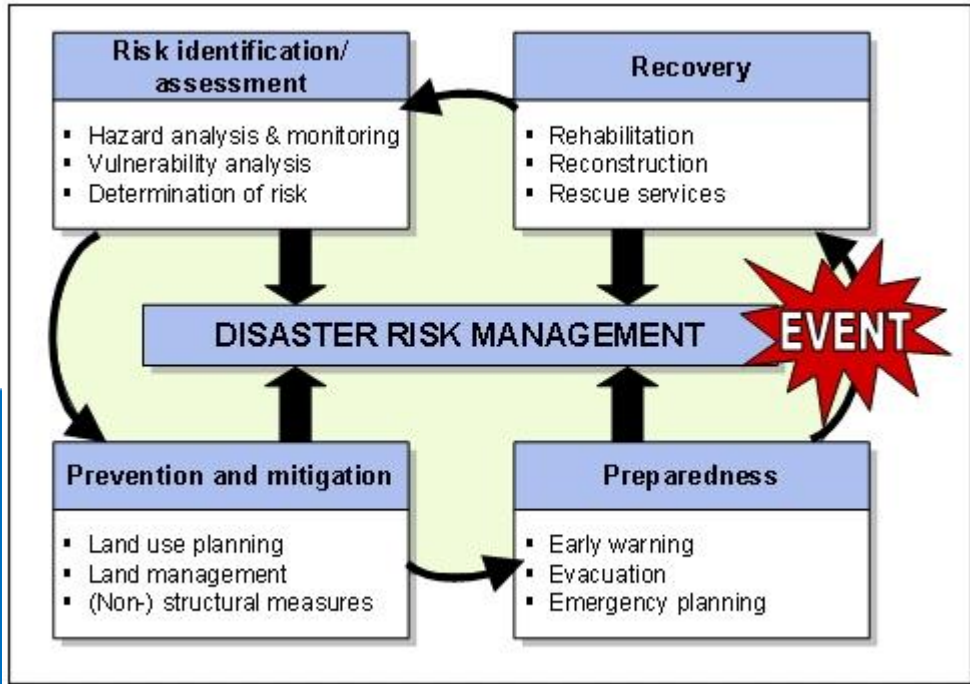
Hazard Identification

- Hazard identification is the foundation of all emergency management activities
- Hazard Profiling is a component of hazard identification where all hazards in a study area are defined
 - General description of hazard
 - Location of hazard and spatial extent of its effects
 - Duration of an event caused by a hazard
 - Seasonal or time-based patterns followed by the hazard
 - Speed of onset of an actual hazard event
 - Availability of warning for the hazard

Class-Activity

Divide into two groups of students.

Prepare a brief hazard profile of any district of Sindh.



Disaster Risk Management

DRM

Disaster Risk Management

- Risk management involves the implementation of the identified risk reduction strategies
- The systematic process of using administrative directives, organizations, and operational skills and capacities to implement strategies, policies and improved coping capacities to lessen the adverse impacts of hazards and the possibility of a disaster
- Decision Support Systems to help decision makers whether to accept, reduce, or minimize the identified risks by the implementation strategies.

The Historical Context

- We do not realize the value of emergency management until we have a major disaster in our community
- The current disaster-prone environment we live in has increased the potential of floods, earthquakes, cyclones, hurricanes, wildfires, and terrorist attacks
(Agree or not?)

Historical Context: Early History

Has ancient roots

- At different time in history the concept of disaster management have been applied differently.
- Initially, whenever there were disasters, individuals and communities tried to do something about them.
- But organized attempts at dealing with disasters did not occur until much later in modern history.

The Historical Context

Emergency management was once all about response to disasters and social and economic recovery

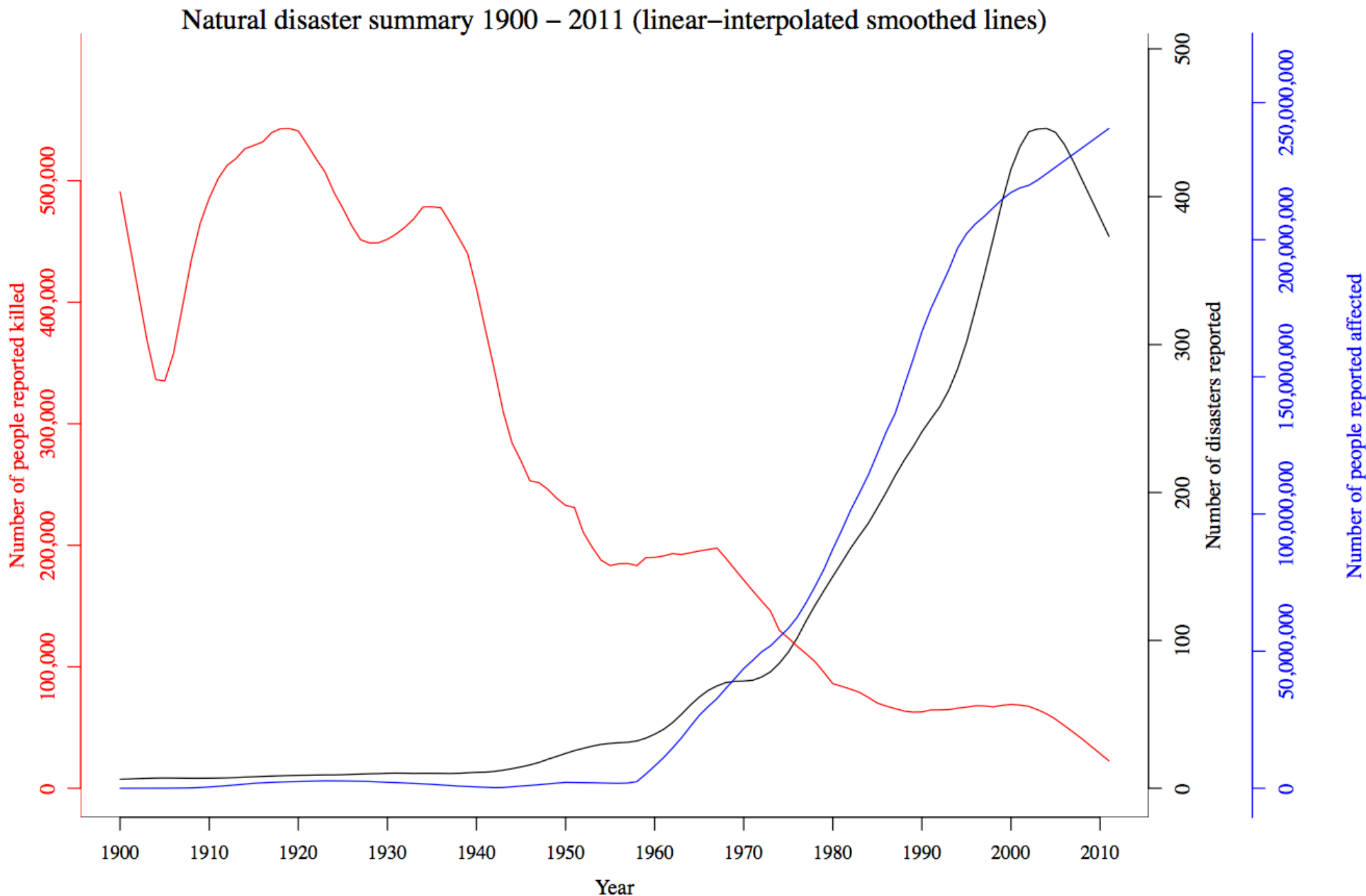
- Now, it also focuses on risk analysis and disaster reduction and mitigation
- Disaster management is now a proactive and multidisciplinary approach

The multidisciplinary nature of DRM?

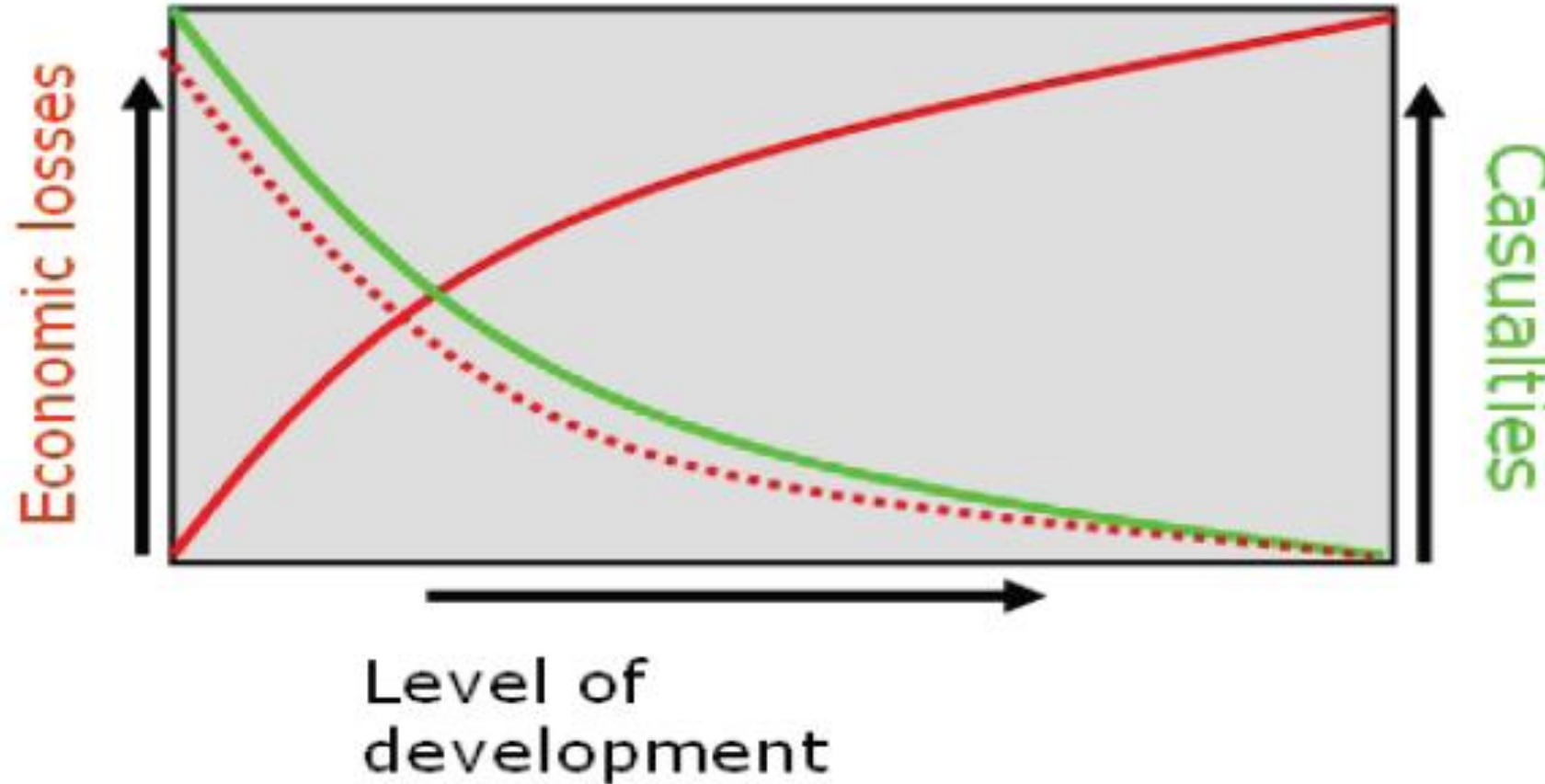
Disasters are Increasing Worldwide?

- Why?
 - Increase in vulnerability
 - Factors leading to higher number of hazardous events.
 - Examples???

DISCUSS



Discuss!!!



Red dotted line: economic losses in relative terms (% of GDP)

Increased Vulnerability

- Factors:

- Rapid increase of the world population.
- Large urbanization rate (related to the population pressure).
- Human settlements in areas that are susceptible to natural hazards (related to population pressure).
- **Most of the world's population lives in hazardous regions, either in coastal zones or in seismically active regions.**
- Development of highly sensitive technologies and the growing susceptibility of modern industrial societies to breakdowns in their infrastructure.
- Developing countries with lower insurance coverage.

Frequency of Destructive Events

- Related to atmospheric extremes are increasing.
- Climate change is negatively related to the occurrence of natural disasters.

Risk Reduction

There are three major components to be considered in disaster management (Hazard, Vulnerability, and Capacity)

Therefore, disaster risk can be reduced through systematic efforts of

- reducing hazard exposure,
- lessening vulnerability and
- developing greater capacity to face these disasters

Determinant of Risk

Five elements of a community

1. People
2. Property
3. Services
4. Livelihood
5. Environment

Each of the above has specific vulnerability to specific hazard

Risk Assessment and Analysis

- What is at Risk? - consequence of Hazards
 - Personal Injury
 - Destruction of property
 - Poor building material and maintenance
 - Damage local economy
 - Opportunity Loss: diversion of economic resources to recovery efforts

Risk and Risk Assessment

Definition of risk

- the probability of harmful consequences, or expected losses (deaths, injuries, property, livelihoods, economic activity disrupted or environment damaged) resulting from interactions between (natural, human-induced or man-made) hazards and vulnerable conditions

Definition of risk assessment:

- A methodology to determine the nature and extent of risk by analyzing potential hazards and evaluating existing conditions of vulnerability that could pose a potential threat or harm to people, livelihoods and the environment on which they depend.

(Source UN-ISDR: <http://www.unisdr.org/eng/library/lib-terminology-eng%20home.htm>)

The process of conducting a risk assessment is based on a review of both technical features of hazards such as their location, intensity, frequency/probability and also the analysis of the physical, social, economic and environmental dimensions of vulnerability and exposure, while taking into account of the coping capacities pertinent to the risk

Disaster Management is:

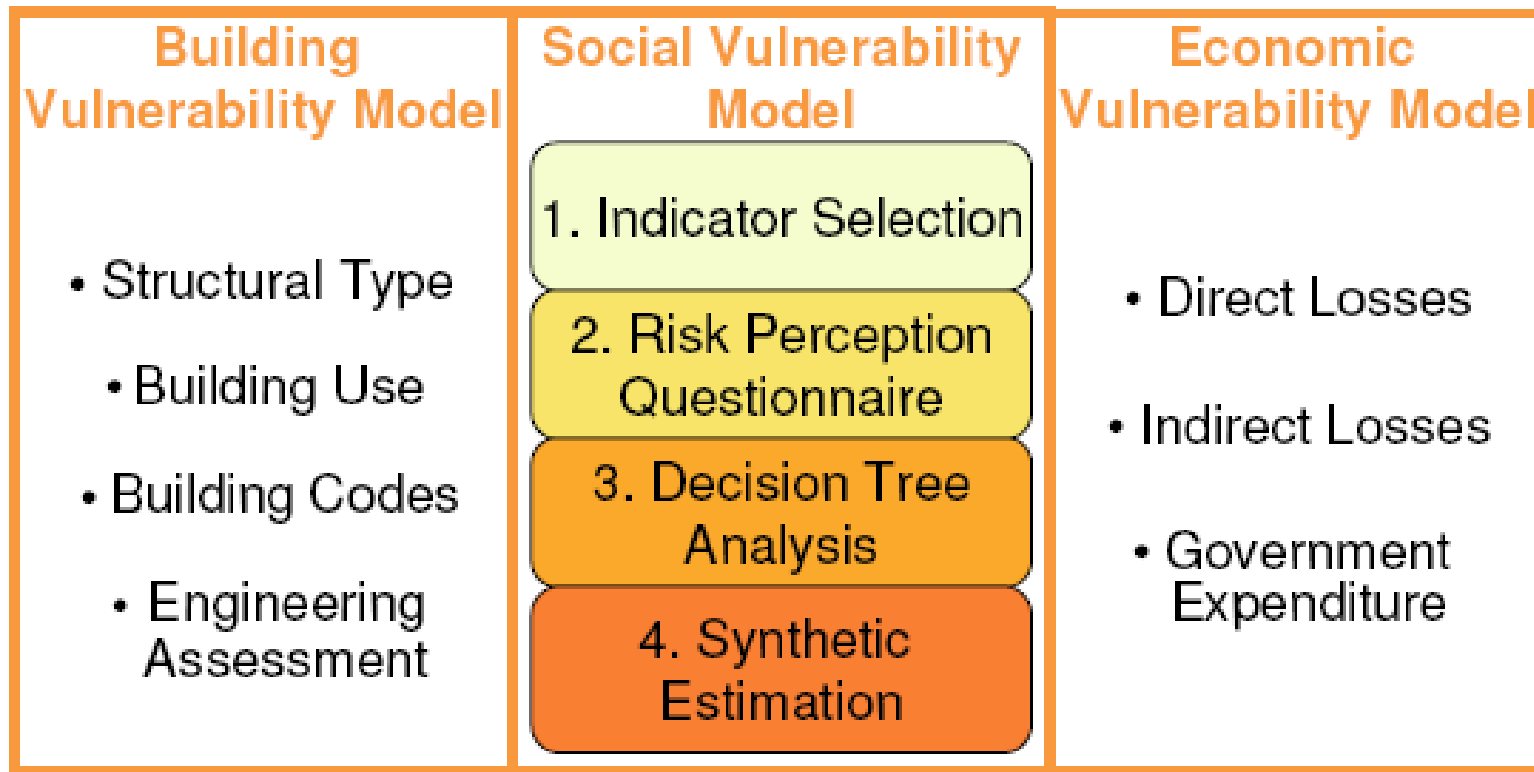
80% generic to all disasters	15% specific to the hazard	5% unique to the event
1. Organisation		
EOC coordination communications transport logistics information reporting	earthquake large numbers of trapped and injured large numbers of homeless and displaced large numbers of dead and missing dead, injured and missing staff damaged critical infrastructure (hospitals) loss of water, electricity, phone networks loss of road, sea, air, rail access long period of SAR, victim extraction high demand for FA, stretchers, triage, medical transport high demand for beds, surgery, blood products, referral high demand for temporary shelter, food, water, energy, clothing, tents, blankets wound infections, amputations, tetanus high demand for orthotics, prosthetics, disability, dental demand for specialised spinal and head injury care high demand for psychosocial support of victims and staff	time place weather geography climate security politics economy governance emergency management capacity logistics capacity leadership solidarity morale corruption
2. Response		
search and rescue evacuation mass casualty management dead and missing security temporary shelter emergency water, sanitation and energy emergency food supplies emergency public and environmental health emergency engineering and public works		
3. Recovery		
lifelines curative and public health care education agriculture trade and commerce	few outbreaks of communicable diseases variable demand for medicines and equipment (acute injury care - high, infectious disease - low, potentially unstable chronic disease - medium)	
4. Rehabilitation and Reconstruction		
people property services livelihoods environment	contamination of water, air and soil toxic chemical and gas leaks/spills urban fires, explosions contaminated, infested and unsafe foods increased vector breeding loss of livelihoods, markets, distribution networks	

Purpose of Risk Analysis

- To plan for public safety by developing
 1. Hazard reduction plans
 2. Vulnerability reduction plans
 3. Emergency Preparedness plans

Risk Analysis Steps

- Identify a Hazard
- Identify the communities exposed to that hazard
- Predict the consequences or impact of the hazard on communities exposed
- Determine the vulnerabilities of the five elements of these communities
- Identify the capacities of the communities to respond to that hazard



Risk Assessment Methodology

- Both Qualitative and Quantitative
- Various computer-based models
- Risk assessment outcomes depend on quality and availability of data

Information required



- **For hazard assessment**
 - Scientific studies/maps, long-term monitoring, historic
 - reports on past incidence of hazards, in particular the
 - location, frequency and severity of the events
 - Understanding of factors influencing hazard occurrence
- **For vulnerability assessment**
 - Hazard characteristics (from the hazard assessment)
 - Characteristics of the element/system of interest:
 - location/site, structural strength/quality, content exposure
 - Environmental quality
- **For risk assessment**
 - Same as for vulnerability assessment, plus specific damage
 - / loss frequency estimates for elements at risk
 - Specific hazard scenario

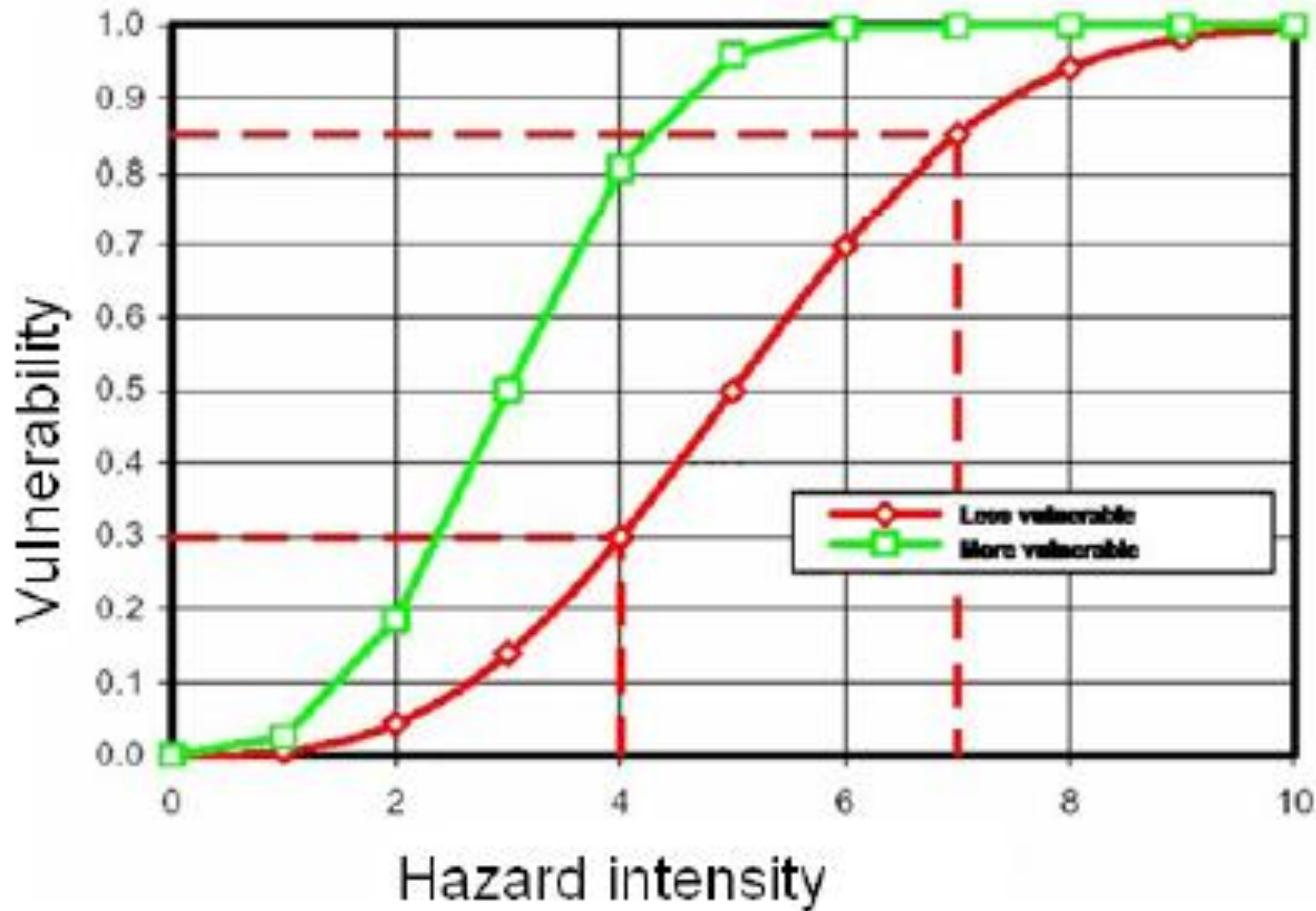
Hazard Risk Assessment

- Process by which threat posed by each identified hazard is investigated
- Risk depends on Hazard likelihood and hazard consequences
- The above two factors are constantly changing as a result of changing environment
- **High-likelihood/high-consequence** hazards are of special consideration
- A comprehensive **Risk Management Program** can greatly reduce life and property losses

Vulnerability Assessment

- Vulnerability indices
- Vulnerability curves
- Fragility curves
- Vulnerability table:

Vulnerability Curve



Risk and Vulnerability Index/indicators

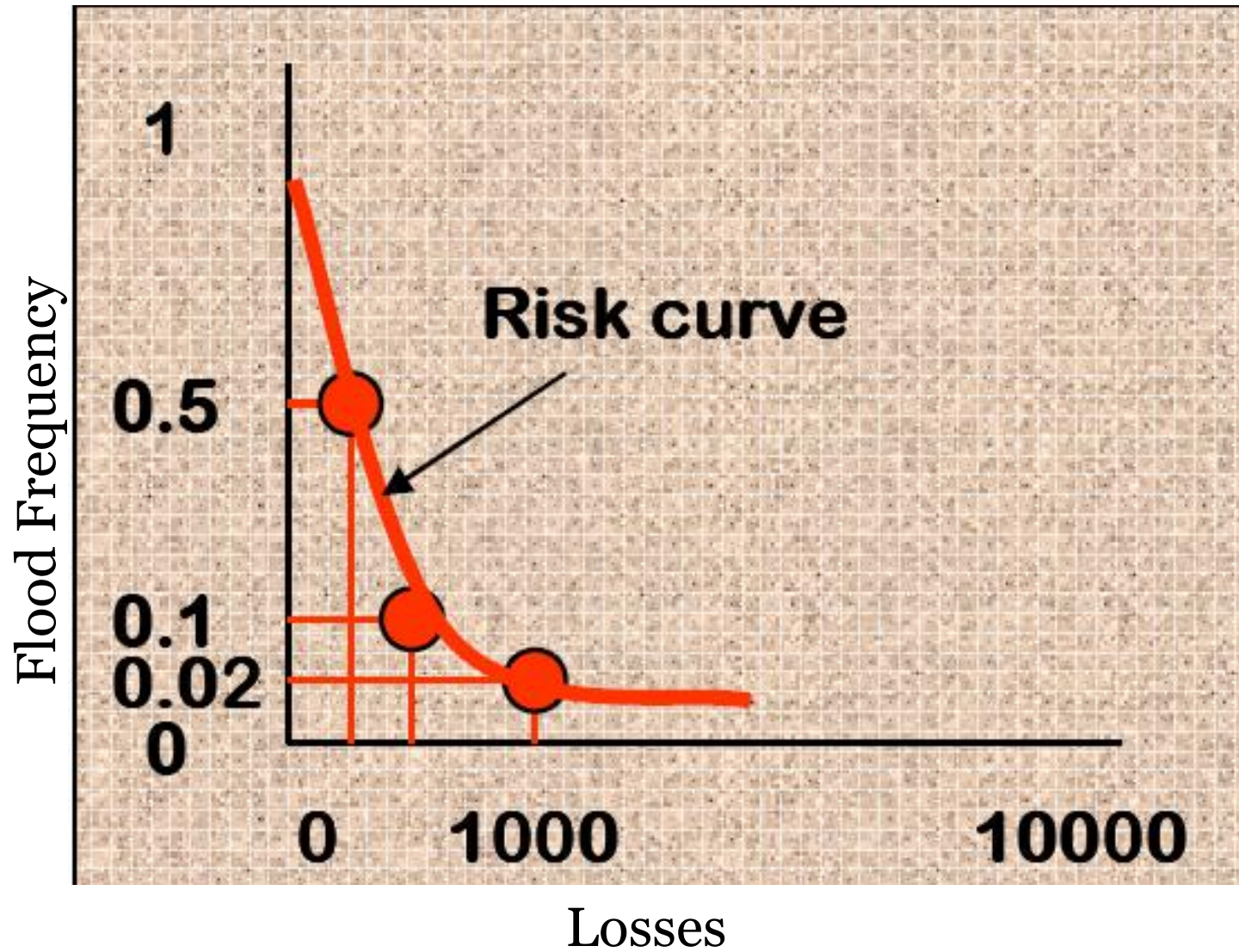
- Various indices defined by
 - World Bank Hotspot
 - UNDP Disaster Risk Index
 - World Economic Forum Environmental Sustainability Indicators
 - International Federation of the Red Cross and Red Crescent Societies (IFRC) World Disaster Reports
 - CRED (Center for Research and Epidemiology of Disasters)
 - More.....

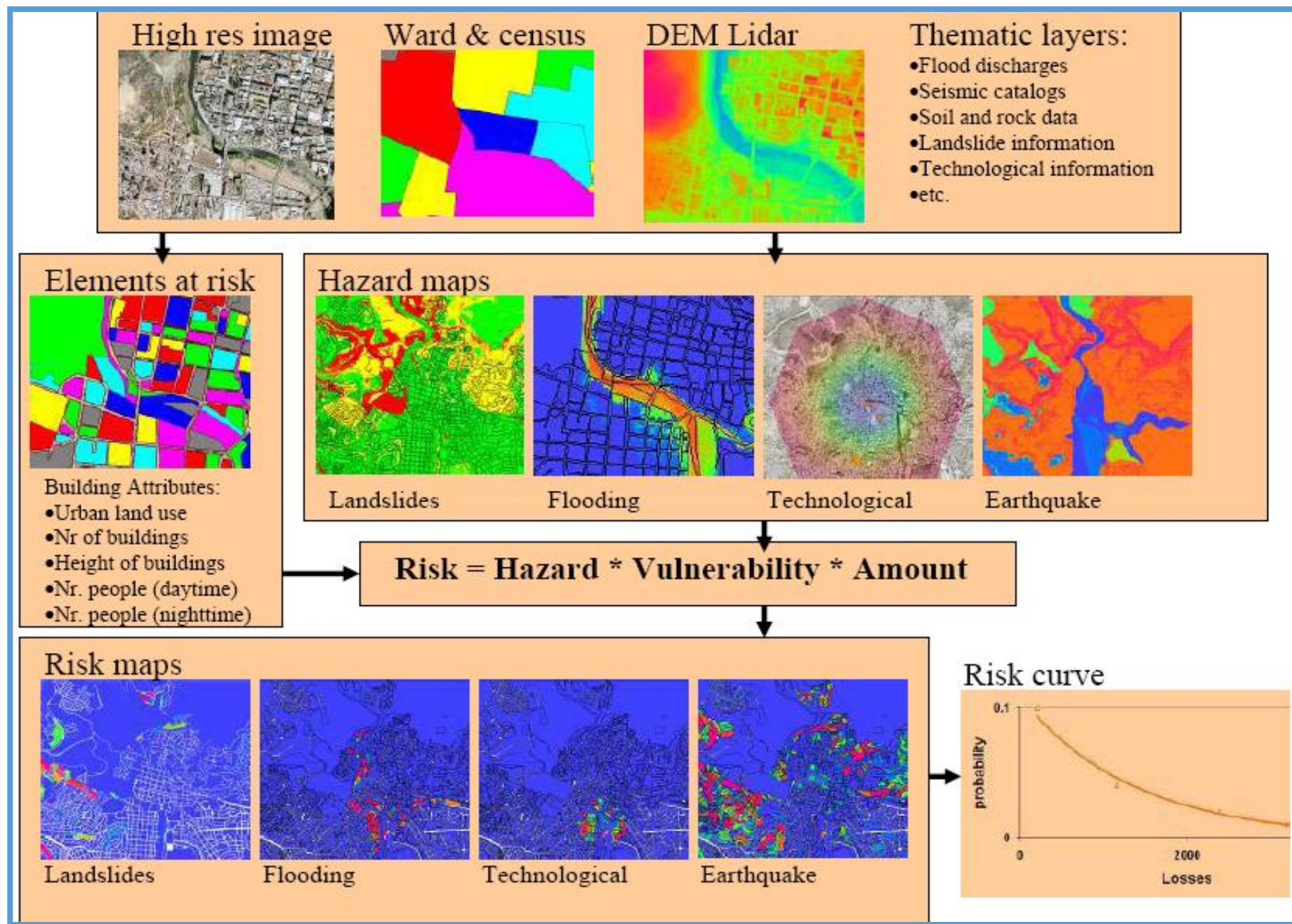
The risk curve



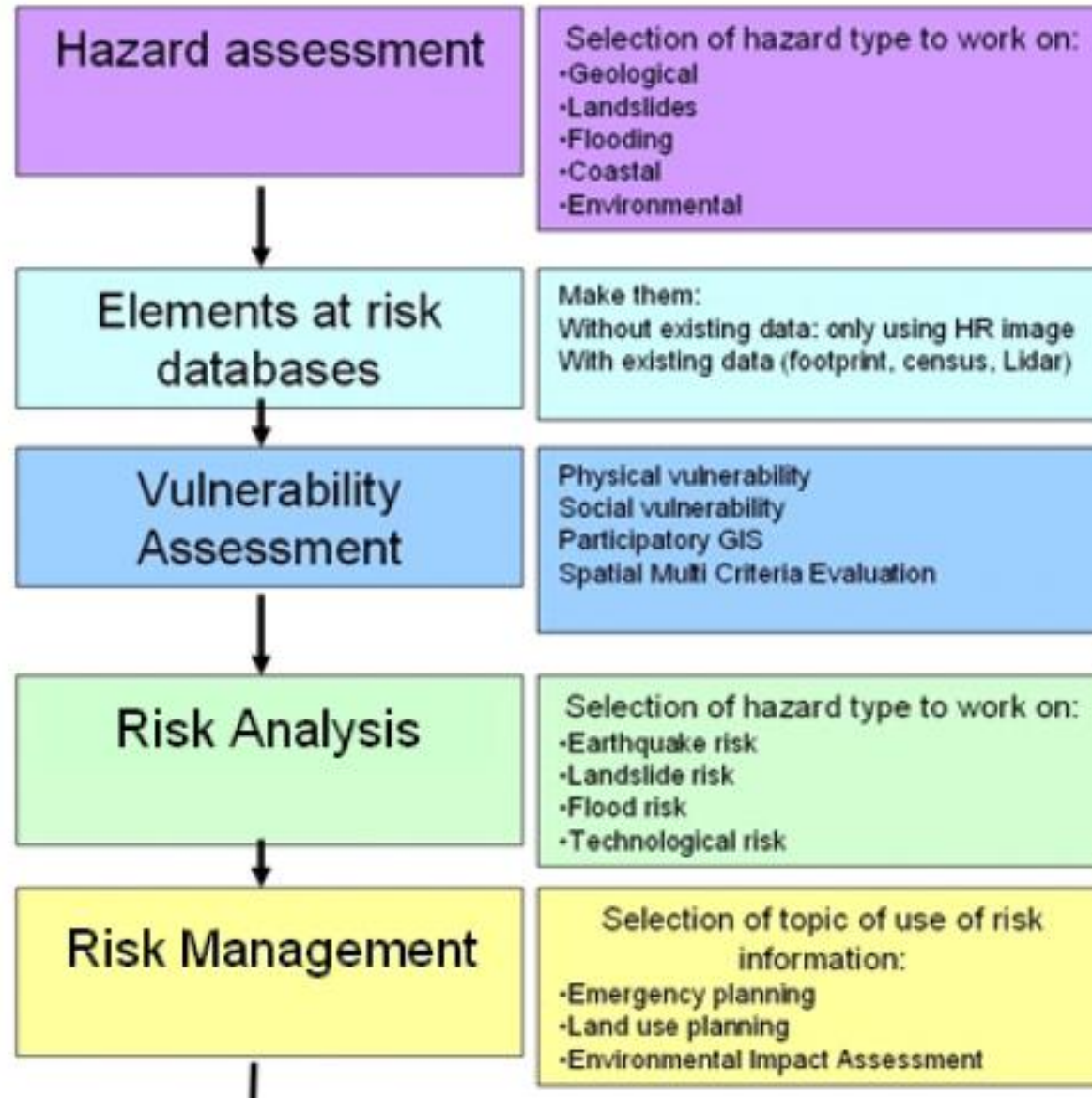
- a graphic representation of the probability of a specific loss associated with a specific return period, such as that expected once every 500 years, 100 years, 50 years, etc
- a complete reporting of potential loss from all possible outcomes from a full spectrum of possible events
- For example,
- a 1% probability of exceedance (1% chance of being larger) is the loss level with a 99% chance of not being exceeded annually, which translates into a 100-year loss return period. Likewise, 0.2% equates to a 500 return period loss

Risk Curve





Disaster Management: New Concepts



Disaster Management Phases

- **Mitigation**—Taking sustained actions to reduce or eliminate long-term risk to people and property from hazards and their effects.
- **Preparedness**—Building the emergency management function to respond effectively to and recover from any hazard.

Disaster Management Phases

- **Response**—Conducting emergency operations to save lives and property by taking action to reduce the hazard to acceptable levels (or eliminate it entirely);
 - evacuating potential victims; providing food, water, shelter, and medical care to those in need; and restoring critical public services
- **Recovery**—Rebuilding communities so that individuals, businesses, and governments can function on their own, return to normal life, and protect against future hazards

Disaster Management Cycle

Critical Phases, Related Activities and Timing

Phase	Time
Mitigation Building codes and hazard/risk zoning, vulnerability analysis, public education	Activities <u>that reduce the effects of disaster before disaster</u>
Preparedness vulnerability analysis, pre-disaster planning, preparedness plans, warning systems	Activities <u>prior to a disaster</u>
Response emergency operations and relief activities, search and rescue, damage assessment	Activities <u>during a disaster</u>
Recovery early recovery (temporary houses), claims processing and grants, medical care and counseling, rehabilitation, reconstruction, post disaster planning	Activities <u>following a disaster</u>

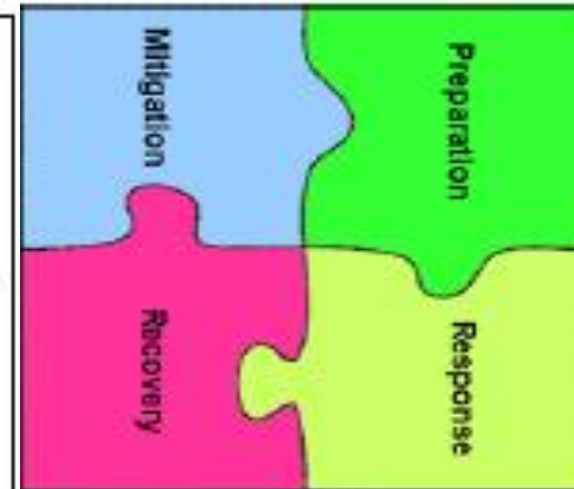
4 Phases



[Saint Paul \(MN\) Department of Emergency Management](#)



[FEMA EMI Course IS-10](#)



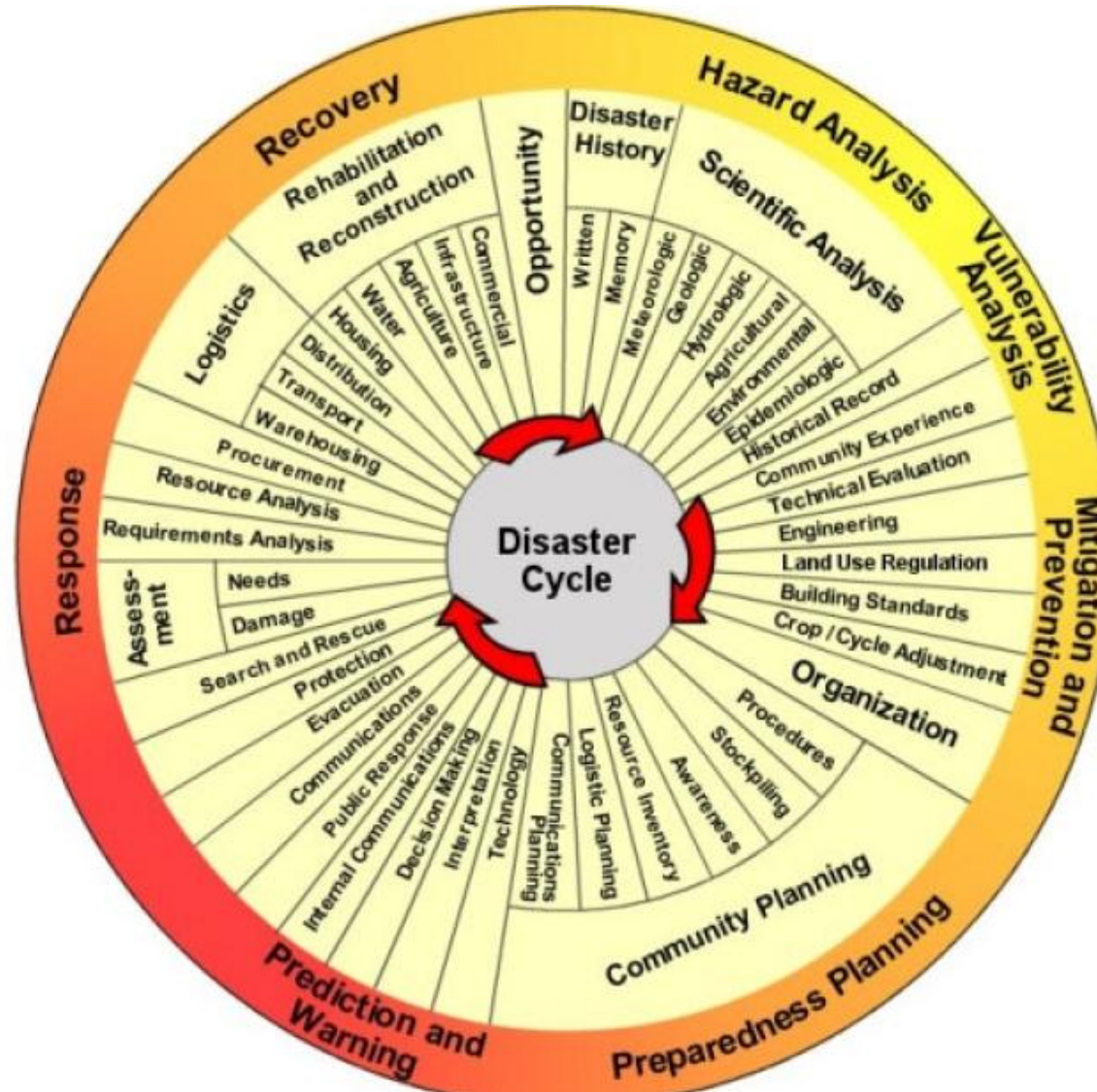
[Arizona Division of Emergency Management](#)



[NCHRP RRD 333 and TCRP RRD 90, A Guide to Planning Resources on Transportation and Hazards](#)



Disaster Management Cycle



Risk Management: Key Elements

Pre-disaster phases				Post-disaster phases	
Risk Identification	Mitigation	Risk Transfer	Preparedness	Emergency response	Rehabilitation and Reconstruction
Hazard Assessment	Physical/structural mitigation works	Insurance/reinsurance	Early warning systems. Communication systems	Humanitarian assistance / rescue	Rehabilitation/reconstruction of damaged critical infrastructure
Vulnerability assessment	Land-use planning and building codes	Financial market instruments	Monitoring and forecasting	Clean-up, temporary repairs and restoration of services	Macroeconomic and budget management
Risk Assessment	Economic incentives	Privatization of public services with safety regulations	Shelter facilities Emergency planning	Damage assessment	Revitalization of affected sectors (exports, tourism)
GIS mapping and scenario building	Education, training and awareness	Calamity funds	Contingency planning (utility companies / public services)	Mobilization of recovery resources	Incorporation of disaster mitigation components in reconstruction

Source: Inter-American Development Bank

Disaster Management, Disaster Risk Reduction, and Risk Transfer

- Risk transfer refers to the process of formally or informally shifting the financial consequences of particular risks from one party to another.
- Disaster risk transfer mechanisms comprise a component of both disaster management and disaster risk reduction.
 - In the former case, financial provision is made to face up to the impacts and consequences of disaster once this materializes.
 - In the latter case, adequate insurance premiums, for example, can promote and encourage disaster risk reduction measures in the insured elements.

Spectrum of Disaster Management Actions

Pre-Incident



Incident



Post-Incident



Prevention

Preparedness

Response

Recovery

Reading Assignment

- Introduction to International Disaster Management
By Damon P. Coppola, Third Edition 2015
 - Chapter 1 & 2 (link is provided)
- Westen Guide Book 1

Discussion/Comment/Question

